Calvin Rutherford

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Education

Purdue University Northwest Hammond, IN
Bachelor of Science in Computer Science *May 2026*
Relevant Courses: Data Structures, Algorithms, Computer Graphics, Operating Systems, Computer Architecture, Network Programming, Java OOP, Database Systems, Intro to AI

Technical Skills

Languages: Java, Python, SQL, TypeScript, HTML/CSS
Technologies: Django, Spring Boot, REST APIs, PostgreSQL, Docker, AWS
Tools: Dokku, Heroku, Linux, Ubuntu, WSL, Nginx, Gunicorn, Slack, Git, GitHub, Jira, Confluence
Concepts: Pandas, NumPy, Scikit-Learn, Logistic Regression, Feature Engineering, NLP, TF-IDF, Data Analysis, Agile Development, Scrum, SDLC,

Professional Experience

Park My Ride Chicago, IL
Backend Developer *May 2026 – Present*

- Design, develop, and maintain backend services powering a transportation technology platform using Django, Django REST Framework, PostgreSQL, Docker, and AWS-based infrastructure. Work directly with company leadership to translate product requirements into scalable technical solutions supporting future public launch and long-term platform growth.
- Develop and maintain 25+ REST API endpoints supporting user management, authentication workflows, account operations, reservation management, administrative functionality, reporting, and business-critical platform services. Implement serializers, viewsets, permissions, validation layers, and business rules to ensure data consistency and system reliability.
- Design relational database models, migrations, indexes, constraints, and query strategies supporting thousands of anticipated user interactions while maintaining data integrity, scalability, and maintainability across evolving business requirements.
- Contribute to deployment and infrastructure workflows using Docker, Dokku, Linux environments, environment variables, PostgreSQL administration tasks, application monitoring, logging, and release management processes.

Technical Projects

Healthcare Workflow Platform | *Python, Django, React, PostgreSQL*

- Designed and maintained data models supporting healthcare-related operational workflows while ensuring data consistency and maintainable application architecture.
- Built secure REST APIs, role-based access, and real-time patient charting with shift summaries to reduce documentation burden and reduce shift handoff errors.

Automated Economic Data Pipeline | *Python, REST APIs, CLI Tooling, Data Engineering, Git, Linux, numpy*

- Built a reproducible Python data acquisition pipeline aggregating 30+ U.S. economic and inequality indicators from FRED, BLS, Census, OECD, and WIID for downstream machine learning analysis.
- Produced raw datasets, normalized time series, and metric-level metadata to support deep learning models predicting income inequality (Gini coefficient).

Imagined Speech Neural Signal Processing Pipeline | *Python, REST APIs, CLI Tooling, Numpy*

- Designed a conceptual machine learning pipeline focused on decoding imagined speech from neural activity through signal preprocessing, feature extraction, classification modeling, and closed-loop feedback concepts.
- Researched approaches involving EEG and intracranial neural signals, artifact reduction, spectral analysis, time-series processing, dimensionality reduction, and classification architectures.

AI Mood Detection | *Python, REST APIs, TF-IDF,Numpy*

- Built a machine learning application capable of classifying emotional sentiment from written text using TF-IDF vectorization, feature engineering techniques, and Logistic Regression models.
- Designed preprocessing pipelines responsible for tokenization, normalization, feature extraction, model training, validation, and prediction workflows.